José Manuel Robles-Aguilar

M.Sc. in Statistical Physics · Artificial Intelligence · Computational Biophysics

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Education

Cinvestav *Mexico City, MX*

M.Sc. in Physics *01 Sep 2023 - 16 Dec 2025*

- **Thesis:** AI-driven Prediction of Circular Dichroism Spectra of Proteins.
- **Advisors:** Dr. Mauricio Demetrio Carbajal-Tinoco & Dr. Damián Jacinto-Méndez
- **Focus:** Integration of Deep Bayesian Neural Networks with the theoretical KCD method.
- **GPA:** 9.1/10

National Polytechnic Institute (IPN) *Mexico City, MX*

B.Sc. in Physics and Mathematics *01 Sep 2017 - 01 Jul 2022*

- **Focus:** Stochastic Processes and Materials Science.
- **GPA:** 8.64/10

Research Interests

Computational Biophysics, Force Fields, MD, AI-driven scientific Discovery, Drug and Materials Discovery, Protein Design, Statistical Mechanics of Soft Matter.

Research Experience

Graduate Researcher *Sep 2023 - Dec 2025*

Dept of Physics, Cinvestav, Mexico City, MX

- Developed a hybrid AI-Physics framework to predict protein optical activity, outperforming state-of-the-art methods by 40%.
- Modeled electromagnetic interactions in proteins using the DeVoe polarizability model.
- Implemented Deep Bayesian Neural Networks to quantify uncertainty in spectral predictions, addressing the ”black box” problem in AI-driven biophysics.
- Analyzed complex biological datasets to correlate structural motifs (Alpha-helix/Beta-sheet/Random Coils/Others) with spectral outputs.

Research Intern - Soft Matter Laboratory *May 2024 - Jul 2024*

Cinvestav, Mexico City, MX

- Investigated Brownian motion of nanoparticles within lipid vesicles.
- Designed a computer vision pipeline for particle tracking and trajectory analysis.
- Calibrated high-precision fluorescence microscopy equipment and captured imaging data across varying focal heights.
- Contributed to the understanding of diffusion processes in biological soft matter systems.

Teaching Assistant *Sep 2021 - Mar 2022*

Center for Computing Research (CIC), Mexico City, MX

- Translated complex technical course material into accessible educational animations using Adobe Illustrator and After Effects.
- Taught and prepared materials for courses on LaTeX and technical documentation in English and Spanish.

Research Fellow (BEIFI Grant) *2018 - 2022*

Superior School of Physics and Mathematics (ESFM-IPN), Mexico City, MX

- Conducted experimental research and computational simulations in Materials Science and Stochastic Processes.

Independent Projects

Drug-Protein Binding Affinity Prediction using GNNs *Jan 2026 - Present*

Personal Portfolio

- Developing a Graph Neural Network (GNN) architecture to predict the binding affinity between drug molecules and target proteins.
- Modeling molecular structures as graphs to capture spatial and topological features essential for accurate affinity scoring.
- Utilizing GPU resources for efficient training on large-scale bio-molecular datasets.

Anomaly Detection & Conformational Clustering in MD Trajectories *Feb 2026 - Present*

PERSONAL PORTFOLIO

- Developing an unsupervised learning pipeline to analyze high-dimensional Molecular Dynamics (MD) simulation data.
- Implementing dimensionality reduction techniques (e.g., PCA, TICA) and clustering algorithms to identify metastable conformational states and transition pathways.
- Applying anomaly detection mechanisms to isolate rare events (such as protein unfolding or cryptic pocket opening) often missed by standard RMSD-based analysis.

Conferences and Schools

13th Symposium on Structural Proteomics *Milan, Italy*

POSTER PRESENTER *Oct 2025*

- Presented research on AI-driven CD spectra prediction to an international audience of computational biophysicists and structural biologists.

Physics and Life Days Conference *Paris, France*

POSTER PRESENTER *Oct 2025*

- Presented the Knowledge-based Circular Dichroism (KCD) method to an international audience of physicists.

AI+Science Summer School *Paris, France*

ATTENDEE *Jul 2025*

- Selected to attend intensive training on the application of Artificial Intelligence in scientific discovery.

QMUL-Durham-Leeds Research School *Durham, England*

ATTENDEE *Jan 2022*

- Collaborated with an international team to develop and defend a research proposal addressing Sustainable Development Goals (SDG).

Technical Skills

Machine Learning & AI: Deep Learning, Bayesian Neural Networks (BNNs), Graph Neural Networks (GNNs), Computer Vision, Regression, Clustering, XGBoost, LightGBM, GenAI, Prompt Engineering, Predictive Modeling, Hyperparameter Optimization.

Programming & Tools: Python (PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Optuna, Seaborn, GROMACS, MDAnalysis, NGLView), Git, Linux/Unix, Bash Scripting, CUDA/GPU Acceleration, SQL, Excel, MATLAB, Adobe Illustrator, After Effects, Power BI, LaTeX.

Scientific Computing: Bioinformatics, Molecular Dynamics, Proteomics, ChimeraX, Statistical Mechanics, Monte Carlo Simulations, Numerical Methods, Quantum Computing, Exploratory Data Analysis, Data Science.

Experimental: Fluorescence Microscopy, CD Spectroscopy, XRD, AFM, EM, UV-Vis, Raman.

Mathematics: Bayesian Statistics, Linear Algebra, Calculus, Stochastic Processes, Game Theory, Non-equilibrium Statistical Mechanics.

Languages: Spanish (Native), English (C1, IELTS: 7.5), French (B1), German (A1).

References

- **Dr. Damián Jacinto-Méndez**, Cinvestav, Department of Physics. **email:** damian.jacinto@cinvestav.mx
- **Dr. José Matías Alvarado-Mentado**, Cinvestav, Dep of Computer Science. **email:** matias@cs.cinvestav.mx